

E-Commerce Order Analytics

Project Documentation & Prediction Model Analysis

Abstract: E-commerce businesses generate huge amounts of transactional data every day. Analyzing this data helps understand customer behavior, sales patterns, and order trends. This project combines Exploratory Data Analysis (EDA) with Machine Learning to predict order status using a Random Forest Classifier.

Objective

- Analyze e-commerce transaction data.
- Discover customer and sales patterns.
- Visualize important business insights.
- Predict order status automatically using machine learning.

Dataset Description

Feature	Description
OrderID	Unique order identifier
CustomerID	Customer identifier
Date	Transaction date
Product	Purchased product
Quantity	Number of items purchased
UnitPrice	Price per unit

Feature	Description
ShippingAddress	Delivery location
PaymentMethod	Payment mode
TrackingNumber	Shipment tracking number
ItemsInCart	Total items in cart
CouponCode	Discount code used
ReferralSource	Source through which customer arrived
TotalPrice	Final amount
OrderStatus	Target variable

Methodology

1. Data Collection

Dataset imported using `pd.read_csv()`

2. Data Cleaning

- Missing value analysis
- Duplicate detection
- CouponCode null value replacement
- Data type verification

3. Feature Engineering

Date column converted into datetime format. New features include:

- **Weekday:** Monday, Tuesday, Wednesday...
- **Month:** January, February...
- **Year:** 2024, 2025...

These features help identify seasonal trends.

Exploratory Data Analysis (EDA)

- **Product Distribution:** Countplot used to understand frequently purchased products.
- **Revenue Analysis:** Barplot created to visualize Product vs Total Revenue.
- **Monthly Trends:** Countplot of months to observe sales trends over time.
- **Payment Method Analysis:** Studied popularity of UPI, Credit Card, Debit Card, and Cash.
- **Coupon Usage:** Analyzed frequency of discount code utilization.
- **Quantity & Price:** Histogram with KDE for quantities and Boxplots for TotalPrice outlier detection.

- **Correlation:** Heatmap generated for Quantity, UnitPrice, ItemsInCart, and TotalPrice.

Machine Learning Model

Target Variable

`OrderStatus`

Preprocessing & Model Selection

Categorical columns transformed using `OneHotEncoder()` within a `ColumnTransformer`.

Model: Random Forest Classifier (`n_estimators=100`)

- Handles categorical features effectively.
- Robust against overfitting.
- Captures nonlinear relationships.

Performance Evaluation

- **Accuracy Score:** Measures overall prediction accuracy.
- **Classification Report:** Precision, Recall, and F1-score.
- **Confusion Matrix:** Visualized using Seaborn Heatmap.

Conclusion

The project demonstrates how EDA and machine learning can be combined to gain insights from e-commerce transactions and automate order status prediction. Random Forest provides strong predictive performance and can be extended for real-world business applications.

Future Scope

- Sales forecasting & Customer segmentation
- Product recommendation systems
- Advanced models: XGBoost and LightGBM

- Deployment: Streamlit Dashboard / Flask API
- Real-time order tracking analytics

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